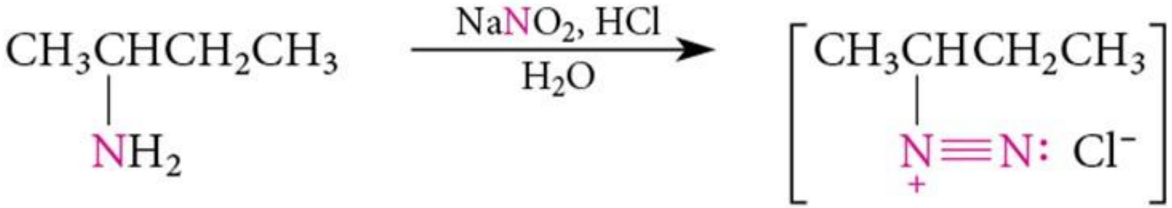
Chapter 23 (Carey Chapter 22)

The Chemistry of Amines

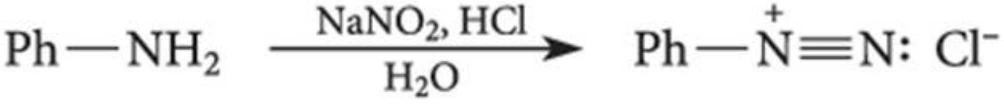
Formation of Diazonium Salts

- ***Diazotization***: The reaction of primary amines with nitrous acid (HNO_2).



2-butanamine
(*sec*-butylamine)

2-butanediazonium chloride
(an aliphatic diazonium salt)
unstable; cannot be isolated

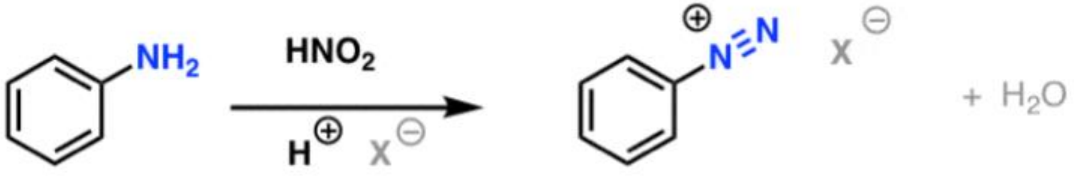


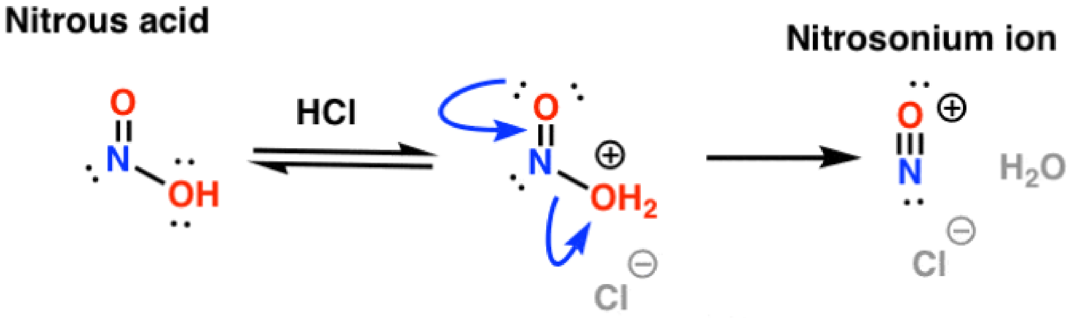
aniline

benzenediazonium chloride
(an aromatic diazonium salt)
can be isolated

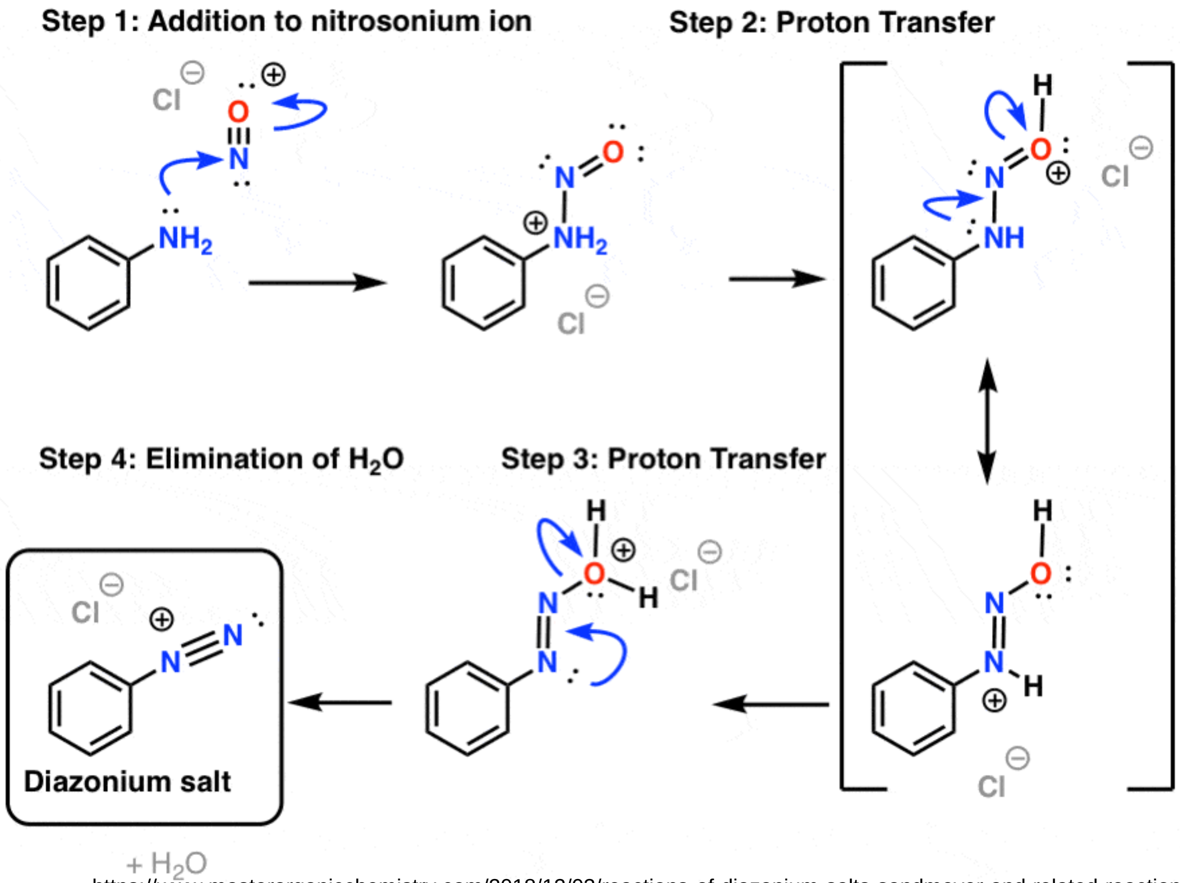
Formation of Diazonium Salts

Formation of Diazonium Salts From Aromatic Amines





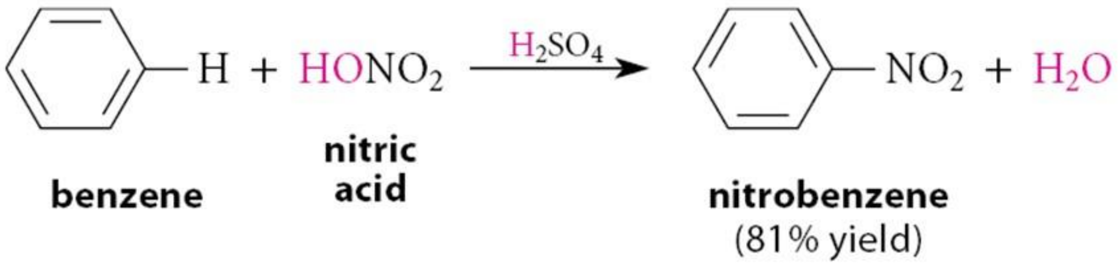
Mechanism: Formation of Diazonium Ions From Aromatic Amines

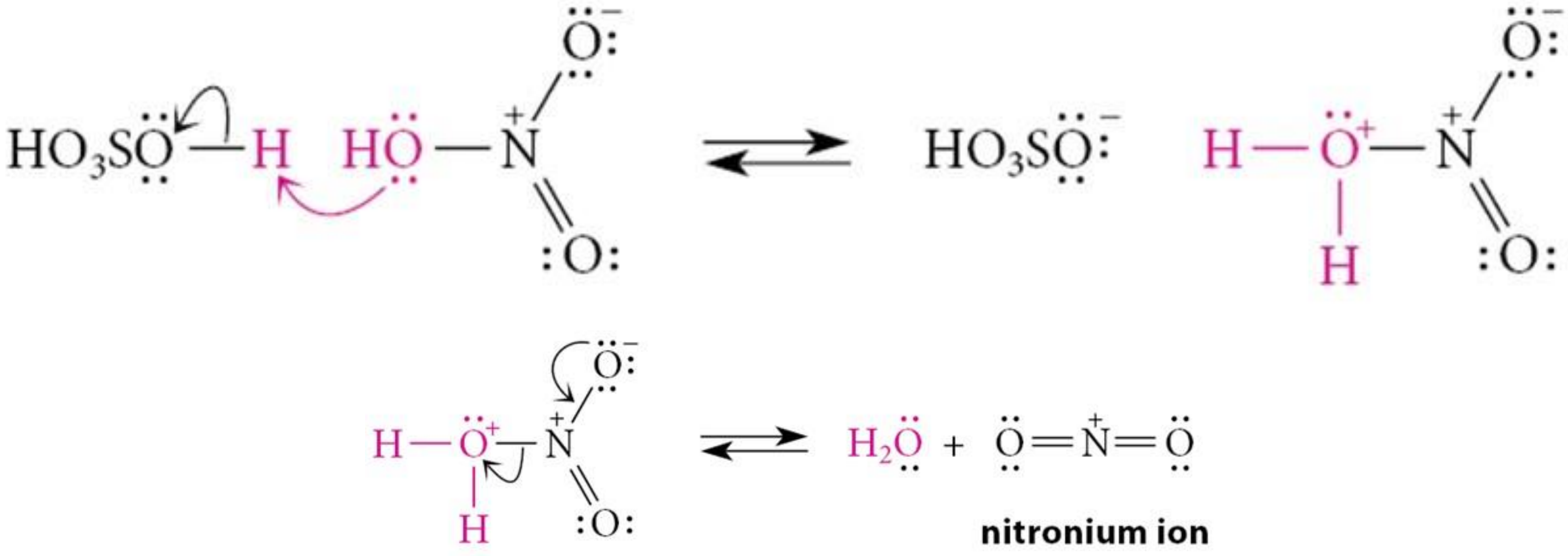


https://www.masterorganicchemistry.com/2018/12/03/reactions-of-diazonium-salts-sandmeyer-and-related-reactions/

Recap: Nitration of Benzene (1 of 2)

- Overall Reaction
- Mechanism
    1. Generation of the electrophile:



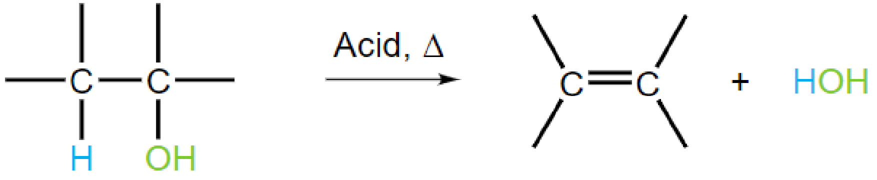


Recap: Dehydration reaction

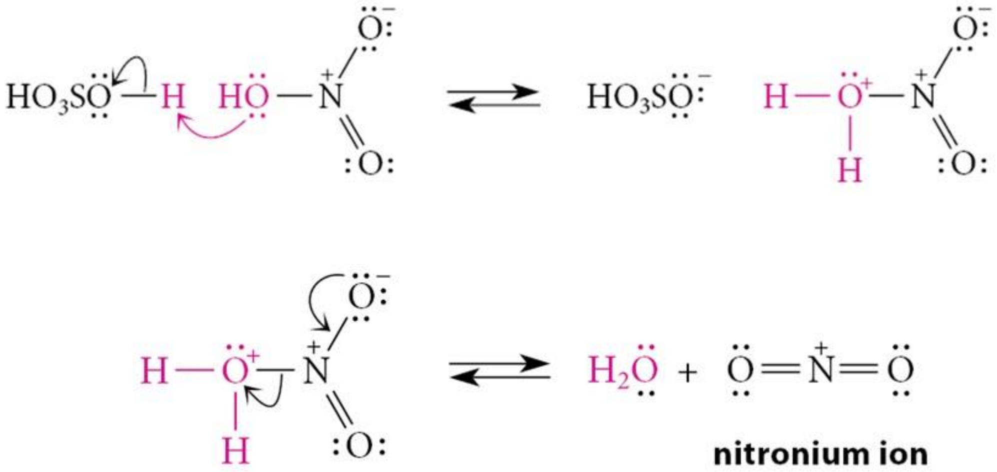
- Acid causes elimination of water from OH compounds

CHEMISTRY COACH | General Chemistry 1 | General Chemistry 2 | General Chemistry 3 | Organic Chemistry

Alcohol dehydration:



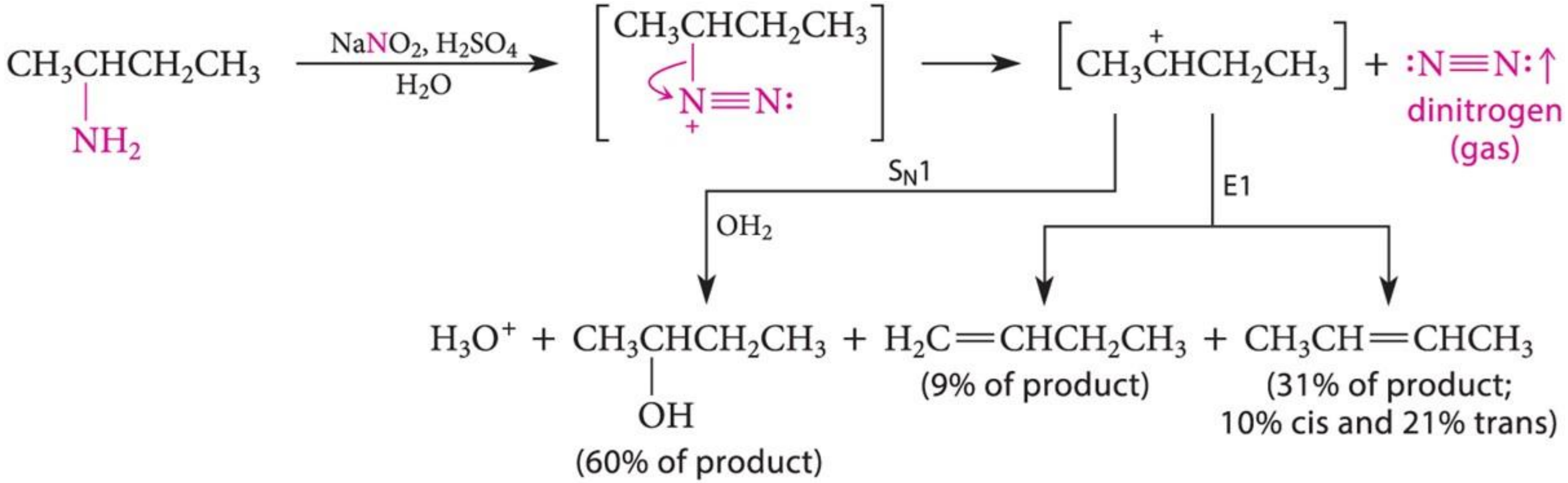
Order of Reactivity: primary < secondary < tertiary
Major product: thermodynamic product



nitronium ion

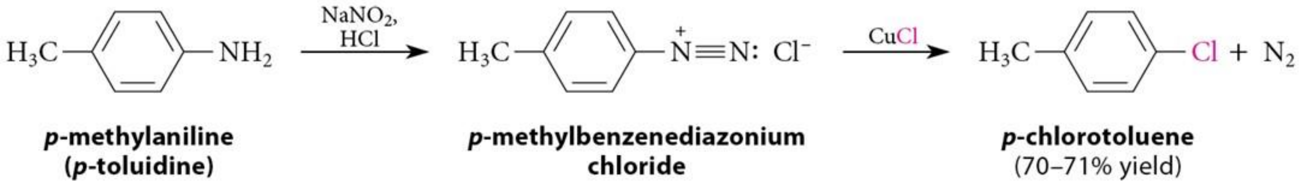
Reactions of Aliphatic Diazonium Salts

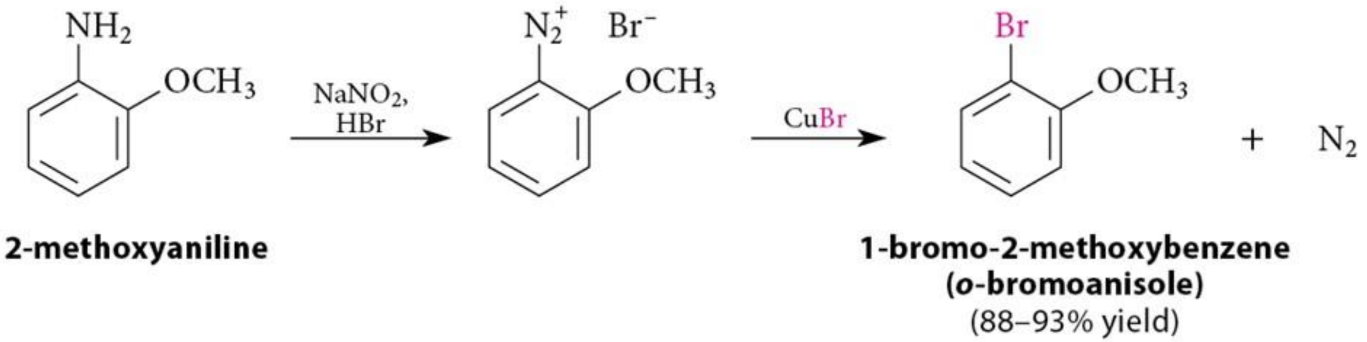
- Diazonium salts possess an *exceptional* leaving group: N≡N
- Aliphatic diazonium salts react immediately to give a complex mixture of products.



Reactions of Aryldiazonium Salts (1 of 3)

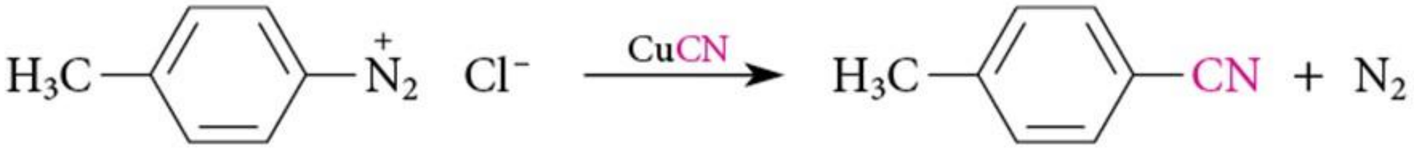
- Aryldiazonium salts are synthetically much more useful.
- The reaction of an aryldiazonium salt with a cuprous salt is called the ***Sandmeyer reaction***





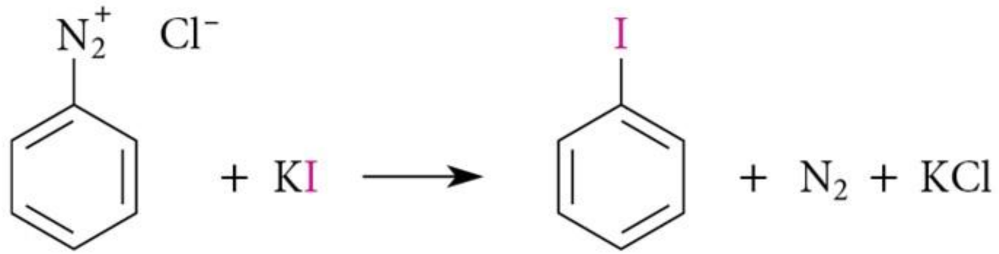
Reactions of Aryldiazonium Salts (2 of 3)

- The diazonium group can be replaced with a cyano group using CuCN.



4-methylbenzonitrile
(67% yield)

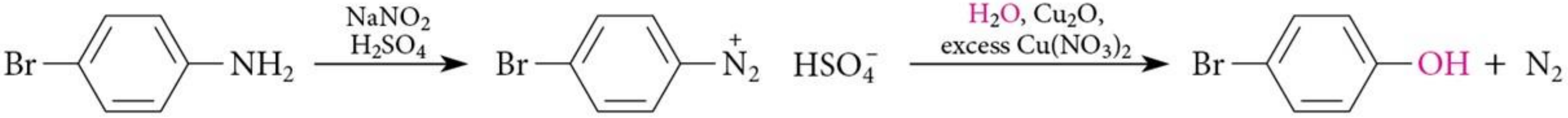
- Aryl iodides can be prepared by reacting the diazonium salt with KI.



(74–76% yield)

Reactions of Aryldiazonium Salts (3 of 3)

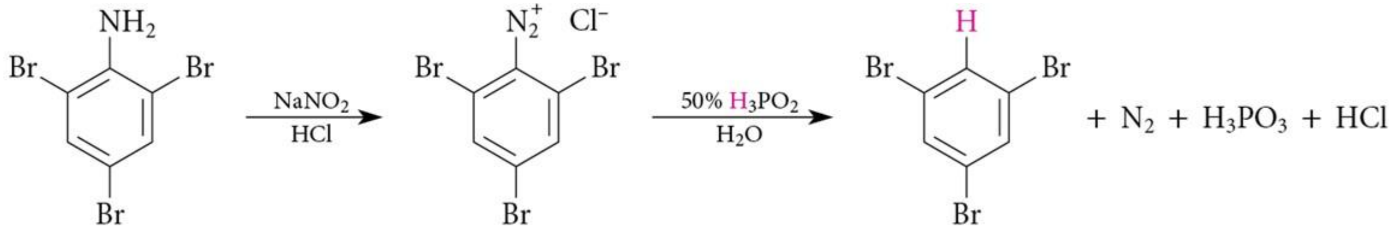
- Hydrolysis to phenols:



***p*-bromoaniline**

***p*-bromophenol**
(95% yield)

- Replacement by hydrogen:



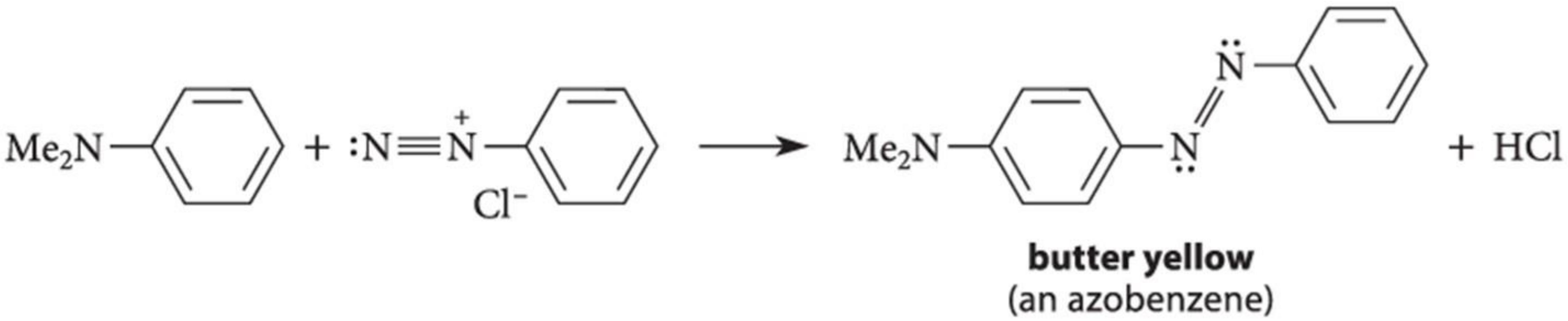
2,4,6-tribromoaniline

2,4,6-tribromobenzenediazonium chloride

1,3,5-tribromobenzene
(70% yield)

Aromatic Substitution with Diazonium Ions

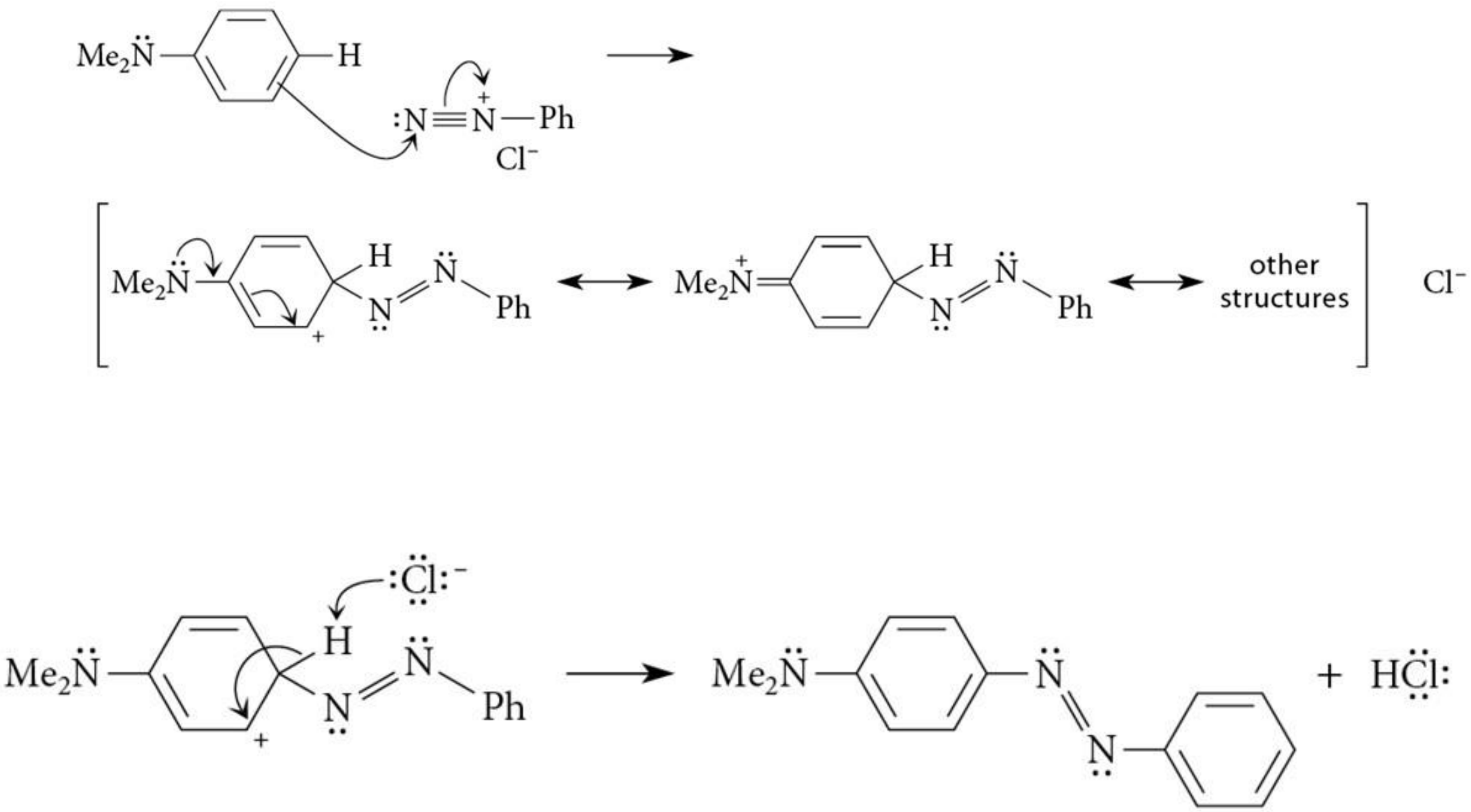
- Activated aromatic compounds can react with aryldiazonium ions to give substituted azobenzenes (Ph—N=N—Ph)



- Comparison of structure/reactivity of diazonium with carbonyl and nitrile.

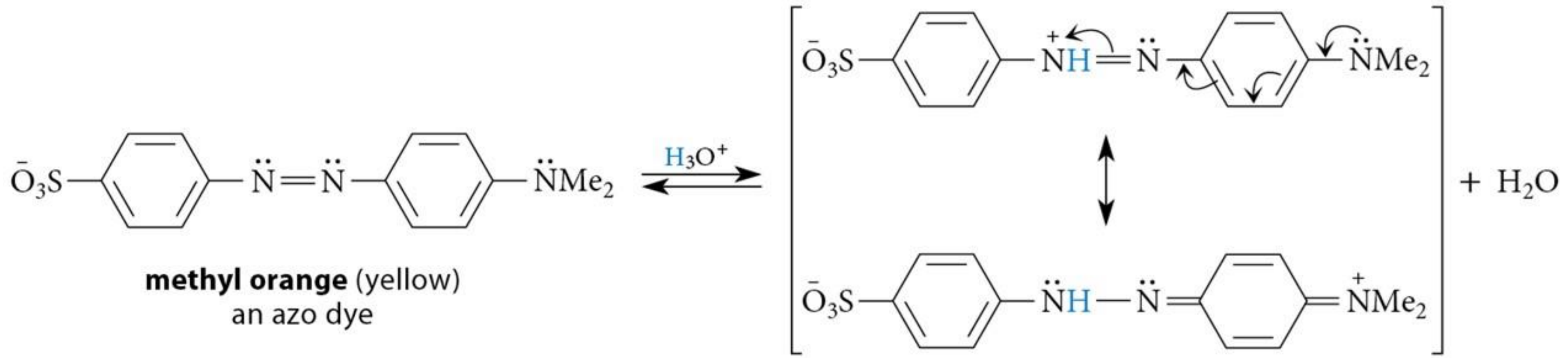
Aromatic Substitution with Diazonium Ions: Mechanism

- The terminal N of the diazonium ion acts as the electrophile.



Azobenzene Derivatives = Azo Dyes

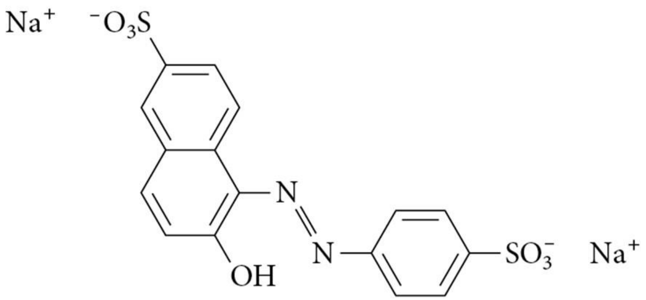
- Azobenzene derivatives (***azo dyes***) are colored due to an extensive conjugated π-electron system.
- Some serve as acid–base indicators.



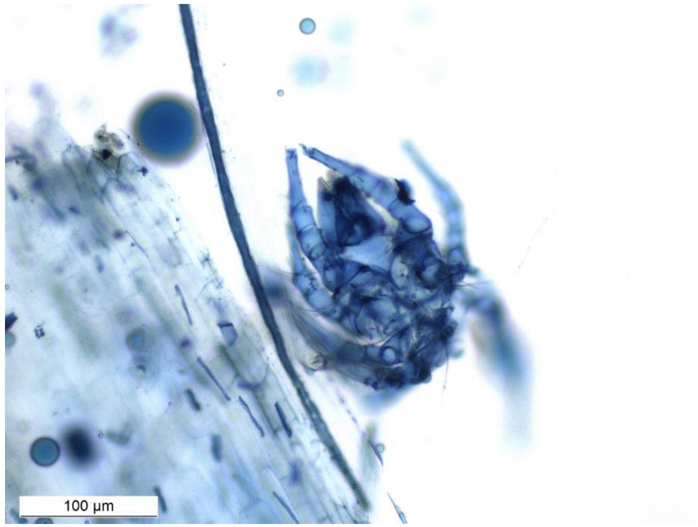
methyl orange (yellow)
an azo dye

protonated methyl orange (red)
pK_a = 3.5

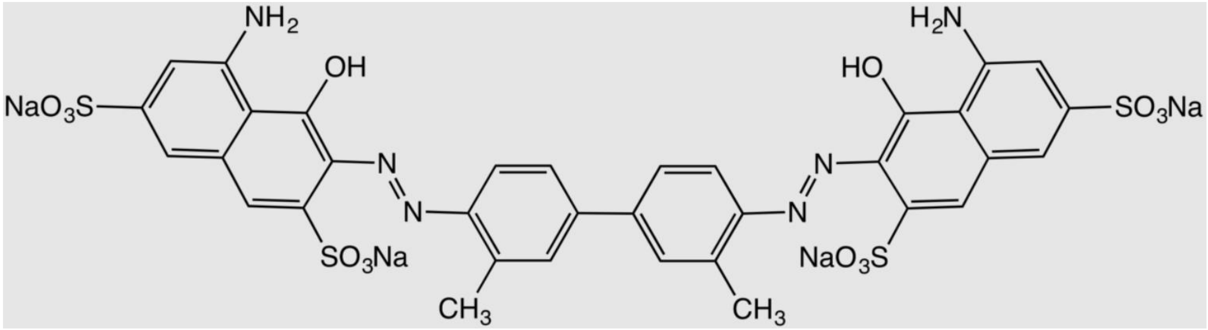
Azo Dyes Uses: Fabrics, Foods, Cosmetics



FD & C Yellow No. 6 ("Sunset Yellow")



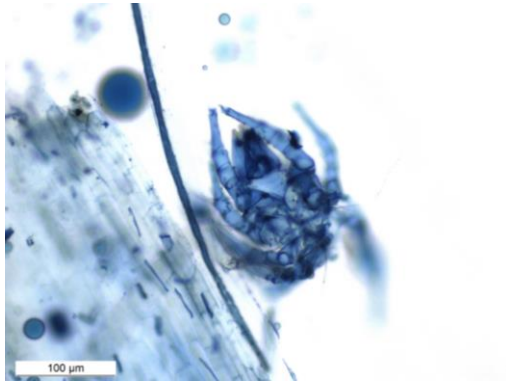
- Trypan blue stains cells

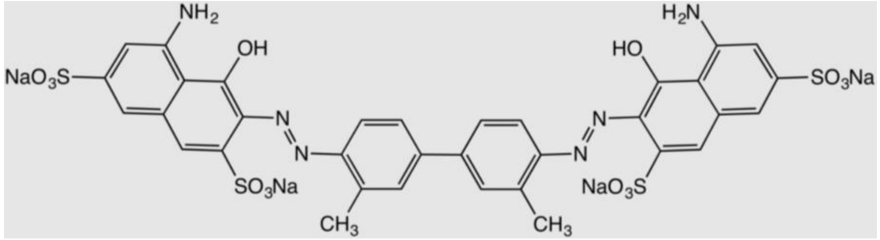


Azo Dyes Uses: Fabrics, Foods, Cosmetics

https://www.youtube.com/watch?v=qfT9uqqme8c

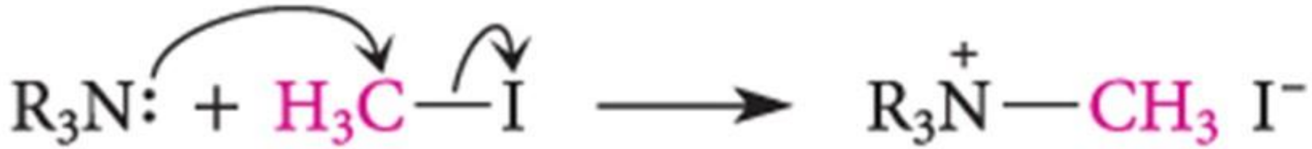
- Trypan blue stains cells





How to prepare primary amines?

- It is difficult to stop the alkylation after the first step:

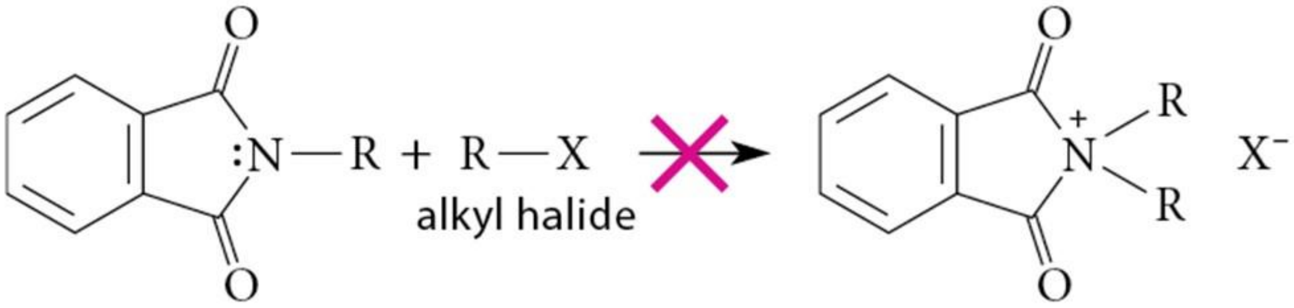


$$\ddot{N}H_3 + CH_3I \longrightarrow CH_3\overset{+}{N}H_3\ I^- + (CH_3)_2\overset{+}{N}H_2\ I^- + (CH_3)_3\overset{+}{N}H\ I^- + (CH_3)_4N^+\ I^-$$

Mono Di Tri Tetra

Synthesis of Primary Amines

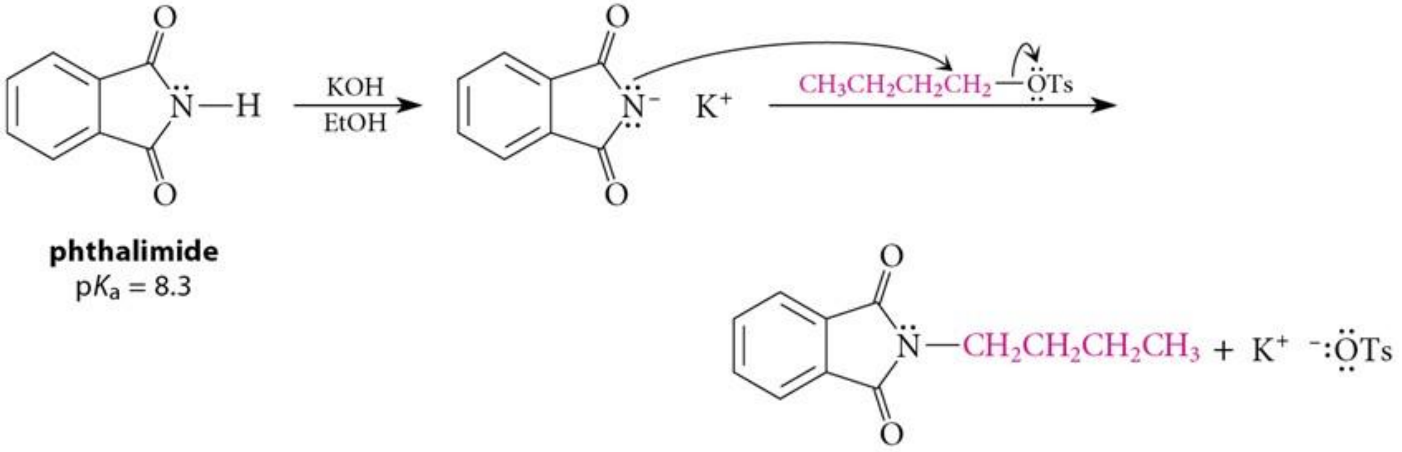
- Direct alkylation is not a good method for the preparation of primary amines.
- The ***Gabriel synthesis*** allows for *controlled* preparation.
  - Alkylation followed by hydrolysis
  - Single alkylation requires a protected amine group, such as phthalimide:



- The reason NH3 would not be the best to make a primary amine shown on the board.

Steps in Gabriel Synthesis

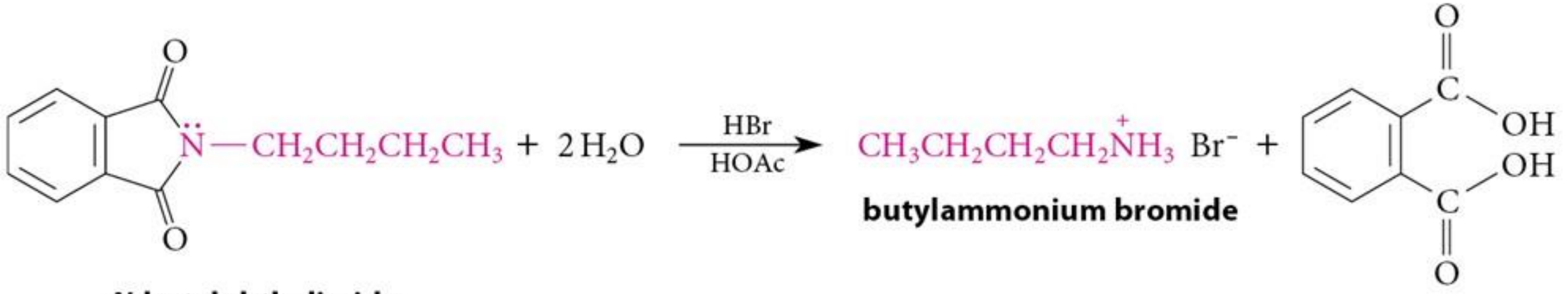
- Alkylation of phthalimide



phthalimide
pK_a = 8.3

***N*-butylphthalimide**

- Hydrolysis of the phthalimide frees the amine.



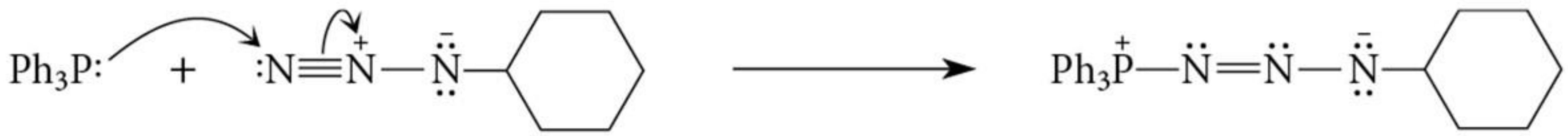
***N*-butylphthalimide**

butylammonium bromide

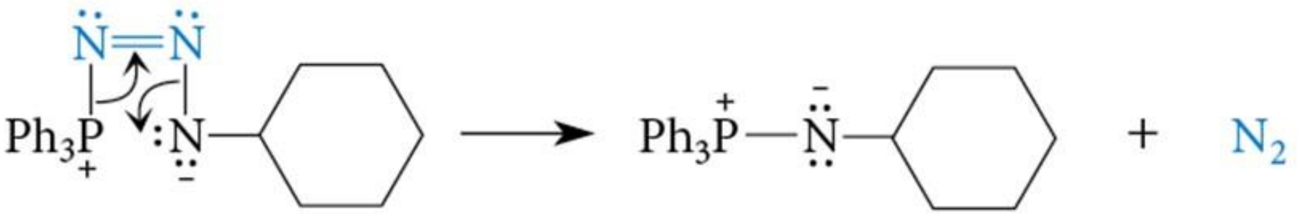
The Staudinger Reaction Also Avoids Multiple Alkylations

$$CH_3CH_2CH_2{-}Br \;+\; Na^+\,N_3^- \xrightarrow[H_2O,\ THF]{} CH_3CH_2CH_2{-}N_3 \;+\; Na^+\,Br^-$$

propyl azide
(88% yield)



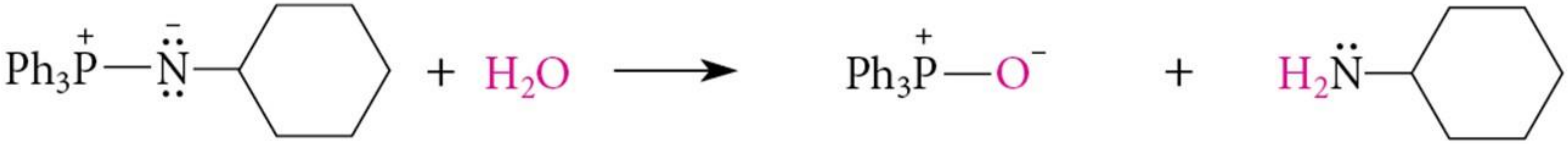
triphenyl-phosphine — an alkyl azide — a phosphazide



phosphazide — an iminophosphorane — **nitrogen gas**

- Wittig reaction recap on the board.

The Staudinger Reaction: Hydrolysis of Iminophosphorane

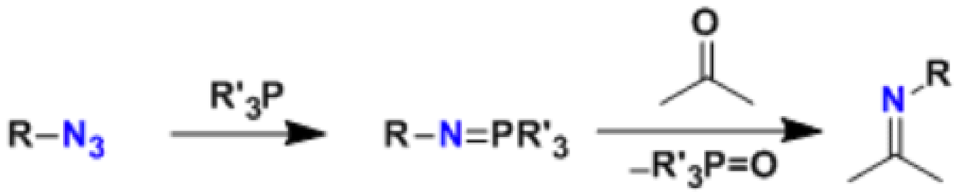


cyclohexyl-iminophosphorane

triphenylphosphine oxide

cyclohexylamine
a primary amine
(95% yield)

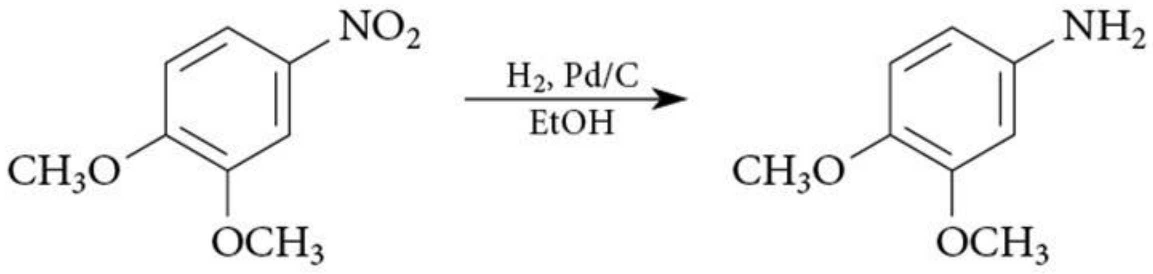
Aza-Wittig reaction



https://en.chem-station.com/reactions-2/2015/03/aza-wittig-reaction.html

Reduction of Nitro Compounds

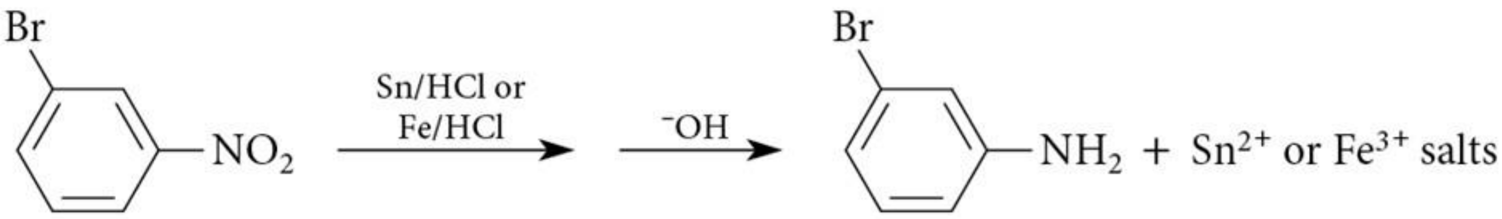
- Catalytic hydrogenation:



1,2-dimethoxy-4-nitrobenzene

3,4-dimethoxyaniline
(97% yield)

- Reduction with finely divided metal powders:



1-bromo-3-nitrobenzene

***m*-bromoaniline**
(80% yield)

Synthesis of aniline by nitro reduction

- Details on the board.

1) Nitration of benzene
2) Reduction of nitro group to amine (aniline)

The directing effect difference between aniline and nitro on the board as well.